

EDUCATIONAL INNOVATION PORTFOLIO

Curriculum Design • Educational Technology • Assessment • Workflow Automation

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Portfolio at a Glance

I am a university English lecturer, curriculum developer, and educational technology specialist with more than seven years of experience in Chinese higher education. My work focuses on making teaching systems more consistent, classroom-ready, and sustainable through curriculum design, assessment improvement, practical automation, and staff technology support.

120 LESSONS REFINED ANNUALLY	100–125 STUDENTS TAUGHT EACH SEMESTER	700–800 PROGRAMME STUDENTS SUPPORTED	7 + 1 INSTRUCTORS + MANAGER USING ATTENDANCE
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Distinctive Contribution

My central contribution is not the use of technology by itself. It is the ability to observe where teachers or students encounter friction, redesign the underlying process, and produce a practical solution that can be used under real classroom conditions.

Qualifications

- Master of Business Administration, University of the People
- Bachelor of Science in Computer Science, University of the People
- Cambridge English Level 5 Certificate in Teaching English to Speakers of Other Languages (CELTA)
- CompTIA Network+
- Google professional certificates in Project Management, Data Analytics, IT Automation with Python, and IT Support

Professional Interests

English for Academic Purposes • curriculum development • educational technology • AI in education • assessment design • learning analytics • programme operations • technology-enhanced learning

1. Design Philosophy

Throughout my career, I have viewed education as a system that can be continuously improved. While teaching remains at the heart of learning, the quality of curriculum, instructional processes, assessment methods, and supporting technology has an equally significant effect on the experience of teachers and students.

Whenever I encounter a process that is inefficient or poorly designed, I instinctively look for ways to improve it. Educational systems should evolve through observation, reflection, testing, and refinement rather than remain static because a process is familiar.

My curriculum design philosophy developed through classroom experience. I observed inconsistent pacing, lessons ending too early, and substantial differences in how complete or usable teaching materials were. Those observations led me to create an eight-stage framework for 80-minute classes and to standardize the supporting teacher guides, answer keys, visual cues, and audio resources around it.

I believe technology should serve education rather than distract from it. I adopt a tool when it solves a genuine instructional or operational problem: reducing repetitive work, making expectations clearer, improving consistency, or giving staff better visibility into what is happening across a programme.

One accomplishment I am especially proud of is creating lessons that require virtually no additional preparation before class. A teacher should be able to open the lesson and begin teaching with the sequence, answer keys, teacher guidance, audio, and student-facing materials already prepared. Reducing teacher workload is not separate from serving students; it allows teachers to devote more attention to interaction and feedback.

Curriculum is never truly finished. Every semester creates new evidence about clarity, pacing, engagement, and usability. Good teaching is not simply about showing up to class. It is about having a well-designed plan that makes students think and learn while giving teachers the tools they need to teach effectively.

2. Project 1 — Eight-Stage Lesson Framework

Role	Curriculum designer and classroom practitioner
Scope	Pre-Intermediate university English lesson set
Status	Implemented and continuously refined
Reach	Approximately 120 lessons per academic year
Core methods	Classroom observation, iterative design, CEFR-aware materials

Overview

The Eight-Stage Lesson Framework is a standardized architecture for 80-minute university English classes. It grew from repeated classroom observation and became the organizing structure for the lesson set I maintain.

Challenge

Early lesson materials varied considerably in pacing, completeness, formatting, and teacher support. Some activities finished before the end of class, forcing teachers to improvise while students were working. Materials also differed in the availability of answer keys, instructions, audio, and visual guidance.

Design Response

I concluded that an 80-minute lesson needed approximately eight stages of about ten minutes each. The framework creates a predictable rhythm among teacher input, student practice, independent work, pair or group interaction, and whole-class feedback. Each stage is supported by teacher guidance and a consistent visual identity.

How It Works

1. Define the lesson outcome and CEFR-appropriate language target.
2. Sequence approximately eight stages to fill the available 80 minutes without rushing or padding.
3. Prepare student-facing activities, teacher guidance, answer keys, visuals, and audio where required.
4. Teach the lesson and observe pacing, instructions, engagement, and student difficulty.
5. Revise the lesson for the next class or semester based on what happened in the classroom.

Operational and Educational Impact

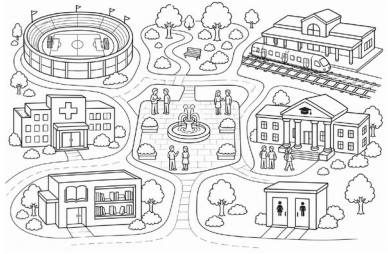
The framework supports approximately 60 lessons per semester and 120 lessons per academic year. It provides teachers with classroom-ready materials and gives students a more predictable instructional flow. Its principal impact is operational and pedagogical consistency; formal comparative learning-outcome data has not yet been collected.

The public case study includes selected pages from the completed 49-page Unit 6 file used by the teaching team and prepared for student printing. The unit contains three lessons, 24 stages, and dedicated teacher-answer pages. No student data is shown.

Selected Evidence

UNIT 6 LESSON 2: City Facilities and Changes

Stage 6.2.1: Picture vocabulary: Find City Facilities



Look at the drawing. Use the word box to name each place.

Word box: stadium • park • railway station • public square • hospital • university • library • public toilet

1. You can watch a football match here: _____
2. You can sit under trees here: _____
3. You can catch a train here: _____
4. You can meet people in an open city space here: _____
5. You can see a doctor here: _____
6. You can study for a degree here: _____
7. You can borrow books here: _____
8. Everyone can use this bathroom: _____
9. Which place do you visit most? _____
10. Why do you visit it? _____

Stage 6.2.8: Proposal writing and speaking: Suggest a New Facility



Your neighbourhood can build one new facility. Compare the five choices.

Choice	Good point	Possible problem
library	quiet study and books	costs money
sports centre	exercise and health	needs space
park	clean air and nature	needs care
market	helps local shops	can be noisy
bus station	easy travel	may add traffic

Write one sentence for each item. Then give your proposal to a partner.

1. Name your choice. _____
2. Say where it should be. _____
3. Give reason 1. _____
4. Give an example. _____
5. Give reason 2. _____
6. Give an example. _____
7. Name one problem. _____
8. Give one solution. _____
9. Use a comparative sentence. _____
10. Finish with a clear recommendation. _____

Stage 1 uses a detailed visual, word bank, and short questions to establish the topic with minimal setup.

Stage 8 moves students into comparison, problem solving, and a supported spoken recommendation.

Reflection and Lesson Learned

The framework was not produced in a single planning session. It emerged from teaching imperfect materials and correcting them while students worked. That direct feedback loop remains central to how I design curriculum. Lesson learned: The best curriculum architecture is specific enough to create consistency but flexible enough to accommodate different language aims and activity types. Classroom observation is the final usability test.

3. Project 2 — Speaking Assessment Platform

Role	Designer, developer, examiner, and maintainer
Scope	Pre-Intermediate one-on-one speaking examinations
Status	Used and reviewed each semester
Reach	Approximately 100–125 assessments personally administered per semester
Technology	HTML, CSS, JavaScript, browser-based delivery

Overview

To improve the consistency and efficiency of individual speaking assessments, I designed a browser-based platform that presents prompts, supports standardized timing, and guides examiners through a consistent sequence without requiring software installation.

Challenge

Speaking assessments are administered individually, with approximately five minutes allocated to each student. Examiners must manage prompts, timing, and sequence while concentrating on student performance. Paper materials add another object to manage and can produce different workflows between examiners.

Design Response

The platform places the required prompts and timed stages into one focused interface. It is intentionally simple so an examiner can begin using it with minimal orientation. Because it runs in a browser, it can be updated between semesters without installing new software on classroom computers.

How It Works

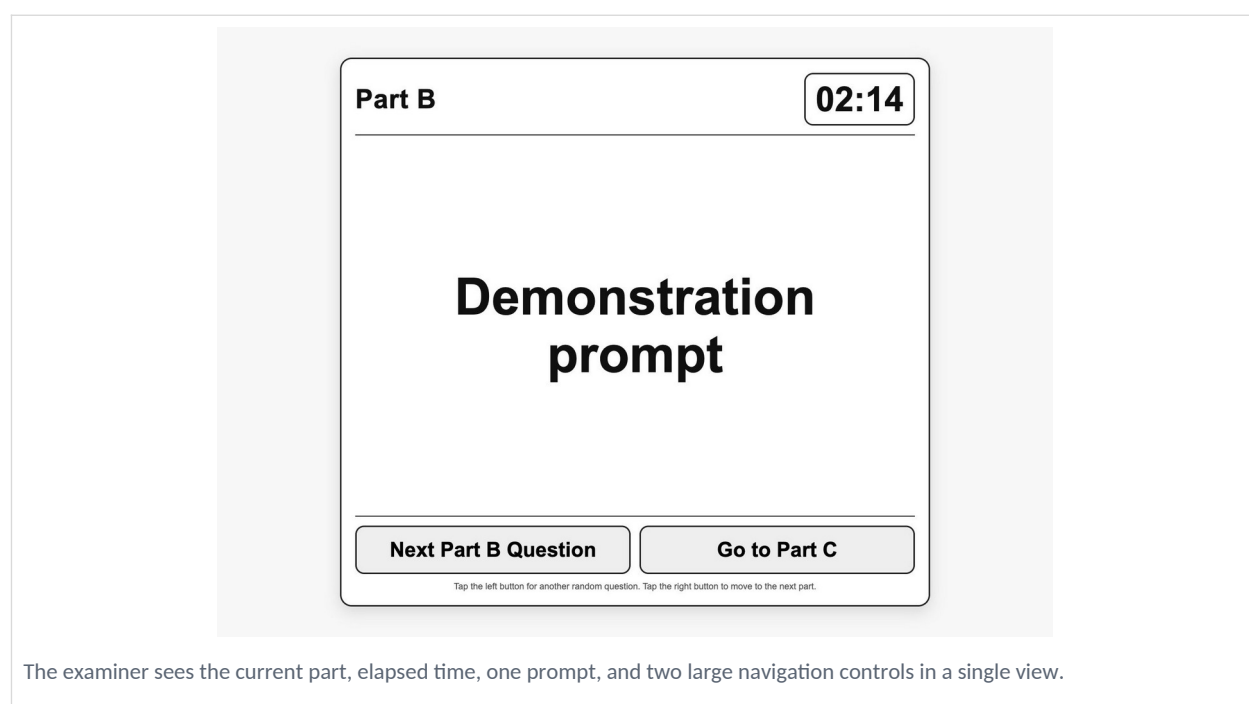
1. Open the assessment in a modern browser.
2. Select or begin the required examination sequence.
3. Present standardized prompts while the interface supports timing and navigation.
4. Keep attention on the student rather than on paper handling or timekeeping.
5. Review the workflow after the assessment period and update it when requirements change.

Operational and Educational Impact

I use the platform for approximately 100–125 five-minute assessments each semester, and two other instructors are confirmed users. It standardizes prompt presentation, timing visibility, and examiner flow while preserving professional judgment. Its present reach is limited to the Pre-Intermediate level, and comparative reliability or time-saving data have not yet been collected.

The public demonstration reproduces the deployed interface while replacing the real question with a neutral label. The live application and source code are linked from waylandchin.com/projects/speaking-assessment-platform/. No student information is stored.

Selected Evidence



Reflection and Lesson Learned

Educational technology is most effective when it quietly organizes routine work. The interface succeeds when the examiner can focus on listening to the student rather than thinking about the software. Lesson learned: For assessment software, reliability, clarity, and ease of use matter more than feature count. Repeated use under real examination conditions provides the best guidance for improvement.

4. Project 3 — AI-Assisted Curriculum Development

Role	Workflow designer, editor, teacher, and quality reviewer
Scope	Student materials, teacher guidance, visuals, and audio
Status	Embedded in ongoing lesson development
Scale	Supports the refinement of approximately 120 lessons annually
Tools	Generative AI, image generation, ElevenLabs, Microsoft Word

Overview

I developed an AI-assisted curriculum workflow to produce stronger materials at a faster pace while retaining human control over language level, pedagogy, accuracy, and classroom usability. AI is used as a production and drafting tool, not as an authority on what students should learn.

Challenge

Initial AI-generated materials often used vocabulary and sentence structures above the students' CEFR level. The outputs also required substantial formatting, sequencing, answer-key development, and testing before they could be taught. Audio and visual resources were inconsistent or absent.

Design Response

I refined prompts and editing practices to constrain student-facing language, then standardized a production workflow around the eight-stage framework. Each stage can include a customized black-and-white line illustration that acts as a visual anchor. ElevenLabs is used for consistent instructional audio when a lesson requires listening material.

How It Works

1. Define the learning outcome, CEFR range, lesson stage, and required student action.
2. Generate a constrained first draft and check vocabulary, grammar, instructions, and cognitive load.
3. Rewrite or regenerate content that exceeds the target level or does not fit the classroom task.
4. Create the student worksheet, teacher guide, answer key, stage illustration, and audio as needed.
5. Teach the material, observe actual student performance, and revise the source lesson.

Operational and Educational Impact

This workflow supports the continuing development and refinement of approximately 60 lessons per semester, or 120 per academic year. A verified production example, Unit 5 Finding Your Way, contains 53 pages, three complete lessons, 24 instructional stages, two original audio files, and a full teacher answer-key section. The demonstrated result is a faster, more consistent production process; I do not claim measured learning gains without comparative data.

Selected pages below come from the final team-accessible Unit 5 file. They demonstrate that AI output passes through CEFR-level editing, instructional design, visual production, audio preparation, formatting, and answer-key development before classroom use.

Selected Evidence

Stage 5.1.2: Listening – Looking for the City Museum



Part A: Listen to the conversation. Circle A, B, or C.

1. Who is looking for the City Museum?

a. Emma and Jake

b. Jake and the shop assistant

c. Emma and the shop assistant

2. Who gives the directions?

a. Emma

b. Jake

c. The shop assistant

3. Where should Emma and Jake go first?

a. Turn left

b. Go straight

c. Cross the road

4. What should they do at the traffic lights?

a. Turn right

b. Turn left

c. Go back

5. Where is the City Museum?

a. Next to the park

b. Next to the train station

c. Next to the bank

Part B: Listen Again and Complete the Sentences. Write ONE word in each blank.

1. My name is _____.

2. This is my friend _____.

3. Go _____ for two blocks.

4. Turn _____ at the traffic lights.

5. The museum is next _____ the park.

6. You can't _____ it.

7. Thank you very _____.

8. Have a _____ day!

An audio-supported listening stage combines an original visual, comprehension questions, and a focused gap-fill task.

Stage 5.1.2: Listening – Looking for the City Museum

Transcript

Narrator: Emma and Jake are visiting a new city. They want to find the City Museum, so they stop at a small shop to ask for directions.

Emma: Excuse me.

Shop Assistant: Hello! Can I help you?

Emma: Yes, please. My name is Emma, and this is my friend Jake. We're looking for the City Museum.

Shop Assistant: Of course. Go straight for two blocks.

Jake: Two blocks. OK.

Shop Assistant: Then turn right at the traffic lights.

Emma: Right at the traffic lights. Thanks.

Shop Assistant: That's right. The City Museum is next to the park. You can't miss it.

Jake: Emma, is it far from here?

Emma: I don't think so.

Shop Assistant: No, Jake. It's only about five minutes on foot.

Jake: Great! Thank you very much.

Emma: Yes, thank you for your help.

Shop Assistant: You're welcome. Have a nice day!

Emma: You too!

Part A

1. A

2. C

3. B

4. A

5. A

Part B

1. Emma

2. Jake

3. straight

4. right

5. to

6. miss

7. much

8. nice

The corresponding transcript and keyed answers allow another instructor to teach without recreating support materials.

Reflection and Lesson Learned

AI became valuable only after I learned to reject plausible but unsuitable output. Curriculum quality comes from the teacher's decisions about level, sequencing, context, and revision—not from the speed of generation alone. Lesson learned: An effective AI workflow needs explicit quality gates. Faster production is useful only when the resulting materials remain accurate, teachable, level-appropriate, and easy for another instructor to use.

5. Project 4 — Centralized Attendance Tracking

Role	Designer, implementer, and day-to-day support
Scope	English programme attendance across three proficiency levels
Status	Used daily
Reach	Seven instructors, the programme manager, and 700–800 students per semester
Platform	Shared spreadsheet workflow with Microsoft 365 / SharePoint

Overview

I designed a centralized attendance workflow so instructors and the programme manager could work from a shared source rather than isolated personal records. The system turns routine attendance entry into programme-level visibility.

Challenge

Attendance had previously been maintained separately by individual teachers. That approach made records difficult to compare, created inconsistent practices, and limited the manager's ability to identify patterns or intervene with students before the end of a semester.

Design Response

The shared workbook organizes attendance by class and instructor while giving the programme manager a consolidated view. I supported access, classroom-computer setup, and the Microsoft 365 / SharePoint workflow needed for staff to use it consistently.

How It Works

1. Maintain a shared roster and class structure for the semester.
2. Give each instructor access through the programme's shared Microsoft environment.
3. Record attendance during normal daily teaching.
4. Review consolidated records to identify recurring absence or other concerns mid-semester.
5. Update the structure when classes, rosters, or programme requirements change.

Operational and Educational Impact

The attendance system is used daily by seven instructors and the programme manager and supports a programme serving approximately 700–800 students each semester. Its main value is timely visibility: the manager can identify problem patterns during the semester rather than reconstructing them from separate records later.

The completed previous-semester workbook consolidates approximately 700–800 students across 180 columns and uses more than 5,000 formulas to apply shared attendance counts and mark calculations. The website presents an anonymized reconstruction; student identities, class assignments, and attendance histories remain private.

Reflection and Lesson Learned

The technical solution was comparatively simple. The difficult and valuable part was moving a team from independent record keeping toward a common habit supported by shared access and daily use. Lesson learned: A system creates value when information becomes actionable. Centralization matters because it supports earlier visibility and intervention, not merely because all records occupy one file.

6. Project 5 — Semester Calendar Automation

Role	Designer, Python developer, tester, and maintainer
Scope	Semester schedules for teachers and classes
Status	Tested macOS application; browser version planned
Adoption	Legacy output used by several colleagues; current app validated
Technology	Python, macOS, Excel, ICS; JavaScript planned

Overview

At the beginning of each semester, teaching schedules arrive as Excel workbooks. I first automated the task with AppleScript, then separated the conversion logic from desktop applications and rebuilt it as a tested Python-powered macOS app. A browser-based JavaScript version is the next planned step.

Challenge

A timetable workbook does not automatically become a usable calendar. Manual entry repeats class, teacher, room, date, and recurrence information and creates opportunities for transcription errors. The workflow also needs to validate imperfect timetable structures without changing a user's calendar automatically.

Design Response

The current app reads XLSX or XLSM files directly, validates dates and timetable cells, previews proposed outputs, and exports recurring ICS calendars into a new timestamped folder. It does not require Excel or Numbers to be open and does not delete or replace Calendar.app calendars.

How It Works

1. Receive the official semester schedule in spreadsheet format.
2. Select the source workbook and worksheet in the macOS app.
3. Enter semester dates, time zone, and any teacher aliases required for validation.
4. Review the proposed teacher, class, event, and warning counts before export.
5. Export shareable ICS files for review and import into common calendar applications.

Operational and Educational Impact

In an end-to-end run against a real timetable structure, the app produced 28 class files, six teacher files, 84 class event series, and 84 teacher event series with zero validation warnings. All 168 event copies had distinct identifiers. The conversion engine passes 14 automated tests. Broader staff adoption of the new app and the JavaScript version have not yet been measured or completed.

The public case study uses fictional timetable data and documents the validated output, safety checks, and roadmap at waylandchin.com/projects/calendar-automation/. Real schedules, names, rooms, workbooks, and calendar files remain private.

Reflection and Lesson Learned

This project shows how a useful automation can mature. The AppleScript proved the need; the Python version added separation, validation, review, testing, safer output, and portability; the planned web version would make the workflow accessible across more systems. Lesson learned: Automation improves in stages: solve the recurring problem, isolate the transformation, add validation and review, then make the workflow easier for other users to access.

7. Project 6 — Assessment Workflow Modernization

Role	Process advocate, materials editor, and classroom implementer
Scope	Pre-Intermediate objective examinations and revision materials
Status	ZipGrade adopted for the Pre-Intermediate level
Related work	Exam formatting feedback and exam-aligned revision lessons
Tools	ZipGrade, bubble sheets, Microsoft Word, shared digital resources

Overview

I helped modernize assessment operations by advocating for optical mark recognition through ZipGrade, improving exam formatting, and creating revision lessons aligned with programme assessments.

Challenge

Manually marking large numbers of multiple-choice responses consumes teacher time without adding corresponding instructional value. Assessment documents also need clear formatting, answerability, and sufficient review support so students understand the format and language demands before the examination.

Design Response

I advocated for objective examinations to use bubble-sheet response capture in the Pre-Intermediate programme. The programme coordinator remains responsible for creating the exams; my contribution is workflow improvement, formatting support, feedback, and aligned review or revision lessons.

How It Works

1. Review the examination structure and identify items suitable for objective scanning.
2. Format multiple-choice sections for bubble-sheet response capture.
3. Use ZipGrade to scan and calculate objective results.
4. Create classroom revision materials aligned with the examination's language and task types.
5. Collect practical feedback and improve formatting or review materials for the next assessment cycle.

Operational and Educational Impact

ZipGrade is used for Pre-Intermediate objective examinations, reducing the need to mark multiple-choice papers manually. The combined workflow improves operational efficiency while preserving teacher attention for speaking, writing, feedback, and other tasks that require professional judgment. Expansion to other levels should be described only after it is confirmed.

The website presents privacy-safe reconstructions of a practice response sheet and a final-exam review unit. No live questions, answer keys, student scores, or confidential results are published.

Reflection and Lesson Learned

Assessment innovation does not always require a new examination model. Sometimes the highest-value improvement is removing repetitive handling so teachers can spend more time interpreting performance and supporting students. Lesson learned: Automation should be applied selectively. Objective responses are well suited to scanning; speaking, writing, and qualitative feedback still require human attention and professional judgment.

8. Project 7 — Digital Enablement and Faculty Support

Role	Peer technology support and workflow enabler
Scope	Classroom computers, shared files, Microsoft accounts, and teaching tools
Status	Ongoing informal contribution
Users	Members of the English teaching team
Environment	Microsoft 365, SharePoint, OneDrive, Authenticator, mixed devices

Overview

Digital systems create programme value only when colleagues can access and use them. Alongside teaching, I provide practical technology support that helps the team open shared files, use classroom computers, and participate in centralized workflows.

Challenge

The teaching environment includes mixed hardware, institutional classroom computers, personal devices, and access restrictions. Some Chinese Android devices make standard Microsoft mobile workflows difficult. Hardware limitations and unfamiliar account procedures can prevent staff from using otherwise useful systems.

Design Response

I help colleagues sign in to Microsoft 365 and SharePoint, configure classroom access, open teaching files, use Microsoft Authenticator, troubleshoot OneDrive, and work around device limitations. When appropriate, I have repurposed available equipment so a colleague can participate in the programme's shared digital workflows.

How It Works

1. Identify the actual access barrier: account, authentication, file permission, device, software, or classroom hardware.
2. Configure the minimum working path to shared teaching resources.
3. Test access on the device and classroom computer the colleague will actually use.
4. Provide practical follow-up when passwords, authentication, syncing, or file formats interrupt the workflow.

5. Use recurring support problems to improve the underlying instructions or system design.

Operational and Educational Impact

This support enabled broader use of shared teaching materials and made programme systems such as centralized attendance more practical. Because it is an informal contribution rather than a formal IT assignment, the portfolio emphasizes the work performed without assigning an unsupported leadership or administrative title.

The public case study documents the support pathway from account identity and authentication through shared files, personal devices, and classroom use. It contains no colleague names, account details, institutional URLs, authentication information, or private teaching files.

Reflection and Lesson Learned

Technology adoption is often harder than technology creation. A technically sound system can still fail if it assumes hardware, accounts, confidence, or support that users do not have. Lesson learned: Design for the least convenient real environment, not the ideal demonstration environment. Adoption improves when access is tested on the actual classroom devices and supported through the first points of friction.

9. Programme Impact Summary

The projects in this portfolio form a connected operating system for teaching rather than a collection of isolated tools. Curriculum standardization improves classroom readiness; AI-assisted production expands the capacity to maintain materials; assessment tools remove repetitive handling; attendance creates shared visibility; calendar automation reduces semester setup; and staff support makes the systems usable under real conditions.

Consistency: A repeatable lesson framework, teacher guidance, answer keys, visuals, and audio create a common quality baseline.

Readiness: Teachers can enter class with complete materials instead of rebuilding or repairing lessons immediately before use.

Visibility: Centralized attendance gives the programme manager access to patterns that isolated records could not reveal promptly.

Efficiency: Browser-based assessment, ZipGrade, and calendar automation reduce routine handling and repetitive entry.

Adoption: Account setup, troubleshooting, and practical peer support connect the designed systems to the people expected to use them.

Continuous improvement: Classroom use and semester cycles provide feedback for the next version of each lesson or tool.

Claims in this portfolio are limited to verified scope, output, adoption, and operational reach. Future evaluation could add preparation-time samples, approved feedback, usage records, and intervention examples before making stronger outcome claims.

10. Future Vision

My next phase of work would connect the existing systems into a clearer evidence loop: curriculum design, student activity, assessment, attendance, and teacher reflection should inform one another without adding unnecessary administrative work. The following ideas are aspirations, not claims about systems already deployed.

AI-assisted formative feedback

Create guided feedback workflows that support teachers while preserving human review and student privacy.

Learning and attendance dashboards

Combine approved, minimal data into early-warning views that help staff identify students needing support.

OCR-assisted written assessment

Use optical character recognition to organize handwritten responses for review without delegating final judgment to automation.

Interactive speaking practice

Create low-stakes browser activities that help students rehearse timed responses before formal one-on-one assessment.

Curriculum quality and version control

Track lesson revisions, evidence, ownership, and classroom feedback so improvement survives staff changes.

LMS and resource integration

Provide a simpler route from curriculum source files to teacher materials, student access, and approved analytics.

The website is the living version of this portfolio. It will carry new demonstrations, privacy-safe evidence, and project updates as they are completed; this document remains a concise downloadable companion.

11. About and Contact

Wayland Chin is a university English lecturer, curriculum developer, and educational technology specialist from New York City, currently working in Jinhua, Zhejiang, China. He combines university teaching experience with an MBA, a B.S. in Computer Science, CELTA, IT and project-management credentials, and practical experience building systems that improve curriculum, assessment, scheduling, attendance, and staff workflows.

Languages

- English — Native or bilingual proficiency
- Cantonese — Professional working proficiency
- Mandarin — Professional working proficiency

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